

Work Experience

- Simula Metropolitan Center for Digital Engineering (SimulaMet)** Oslo, Norway
Ph.D. Student (Full-time), Affiliated with Oslo Metropolitan University May 2023 – Present
Conducting research on **multimodal learning for robust and trustworthy vision-language models**; supervising master's students in AI and medical imaging; representing Simula at the NORA Student Council and organizing academic and outreach events; participating in Simula Academy outreach programs to inspire and attract master's students to research.
- The Alan Turing Institute** London, United Kingdom
Ph.D. Enrichment Student for the Turing Enrichment Scheme 2024/25 Oct 2024 – Jul 2025
Selected through the Norwegian Artificial Intelligence Research Consortium (NORA); undertook a six-month research stay in London as part of doctoral training; served as Student Representative for the cohort; contributed to discussions that led to the extension of Kvasir-VQA into a multi-task and a multi-complexity benchmark for medical reasoning in gastrointestinal endoscopy; participated in Turing-Roche Co-working sessions, AI-UK 2025 volunteering, and professional development workshops; fostered interdisciplinary collaborations across UK research labs and universities.
- Thapathali Campus, Tribhuvan University** Kathmandu, Nepal
Adjunct Lecturer (Contract-based) Oct 2022 – May 2023
 - Lectures and labs for "A Practical Approach to Deep Learning" course.
 - Supervised MS projects related to deep learning.
- North Star Developer's Village** Lalitpur, Nepal
Engineering Manager Apr 2019 – May 2023
 - ML face verification; pose-based exam cheating prevention; image/audio annotation; ML training platform.
 - Semantic essay ranking; feedback bot; topic modelling/analysis; ML models exposed via APIs.
- Leapfrog Technology** Kathmandu, Nepal
AI Intern Oct 2018 – Mar 2019
 - Predictive models for time-series weather and pollution forecasting (ARIMA and windowing-based).
 - Model deployment and web-based interactive interface.
- Department of Electronics & Computer Engineering (DOECE)** Lalitpur, Nepal
Student Research Assistant for UGC Research Grant Project Jan 2018 – Sep 2018
 - Assisted Prof. Dr. Nanda Bikram Adhikari on algorithm/tool research & implementation.
 - Sensor data storage with GPS traces & data mining. Used RNN, RapidMiner, Hadoop, WebSocket, etc.

Education

- Oslo Metropolitan University (OsloMet)** Oslo, Norway
Doctor of Philosophy (Ph.D.), at the Faculty of Technology, Art and Design (TKD) May 2023 – May 2026
Thesis: Bridging Multimedia Modalities: Enhanced Multimodal AI Understanding and Intelligent Agents.
Coursework: Topics in Artificial Intelligence and Machine Learning; Scientific Research Methods and Data Analysis in Engineering Science; Engineering Science and Ethics; Communicating Scientific Research (Simula Academy); Neural Methods in Natural Language Processing (University of Oslo).
Research Interests: Multimodal AI; Vision-Language Models; Parameter-Efficient Fine-Tuning; Uncertainty Quantification.
- Thapathali Campus, Tribhuvan University** Kathmandu, Nepal
Master's Degree in Informatics & Intelligent Systems Engineering Oct 2020 – Oct 2022
Full Scholarship for Academic Merit, Top of Class
Relevant coursework: Natural Language Processing; Computer Vision; Applied Machine Learning; Deep Learning; Computational Intelligence; Information Theory; Information Visualization; Optimization.
- Pulchowk Campus, Tribhuvan University** Lalitpur, Nepal
Bachelor's Degree in Computer Engineering Oct 2015 – Oct 2019
Full Scholarship for Academic Merit
Relevant coursework: Artificial Intelligence; Probability & Statistics; Data Mining; Big Data; Theory of Computation.

Technical Capabilities

Programming: Python, Android (Java/Kotlin), SQL, Bash, C/C++.

Libraries and tools: ML/DL, data analysis, numerical computing, visualization, web-frameworks, containerization, cloud, and backend-infrastructure ecosystems (e.g., PyTorch, TensorFlow/Keras, NumPy/SciPy, Pandas, Scikit-learn, Matplotlib, OpenCV, Django, FastAPI, HuggingFace, LangChain, vLLM, PEFT/LoRA, Gradio, Streamlit, Git, Docker, Jupyter, etc.).

Languages: English (Fluent), Nepali (Native), Hindi (Advanced), Norwegian (Basic).

Academic Projects

MS Thesis: [AI-Based Soccer Game Summarization: From Video Highlights to Dynamic Text Summaries](#).

Best thesis in the department (by scores). In collaboration with the Department of Holistic Systems, **SimulaMet**.

MS Project: [Deep Learning-based Pose Estimation for Dystonia Score Prediction](#). In collaboration with **NAAMII**.

Bachelor's Thesis: [Assessment of Spatiotemporal Weather and Pollution Proxies with RNN and ARIMA](#).

Best Project Award by Department.

Bachelor's Project: [Human Facial Landmarks Detection using Multi-Layer Perceptron](#).

Selected Publications

2025: [Kvasir-VQA-x1: A Multimodal Dataset for Medical Reasoning and Robust MedVQA in Gastrointestinal Endoscopy](#). *MICCAI Workshop on Data Engineering in Medical Imaging*. LNCS, Springer.

2024: [Kvasir-VQA: A Text-Image Pair GI Tract Dataset](#). *VLM4Bio*, ACM Multimedia.

2024: [SoccerNet-Echoes: A Soccer Game Audio Commentary Dataset](#). *IEEE International Symposium on Multimedia*

2023: [Bridging Multimedia Modalities: Enhanced Multimodal AI Understanding and Intelligent Agents](#). *ICMI*, ACM.

2022: [Soccer Game Summarization using Audio Commentary, Metadata, and Captions](#). *Workshop on User-centric Narrative Summarization of Long Videos (NarSUM)*, ACM Multimedia. (Part of master's thesis work)

Awards & Certifications

NAAMII AI Winter School: [Full scholarship for participation](#). Topics: Deep Learning; Computer Vision; Statistics & Algorithms; Data Science; Optimization.

Hult Prize TU Winner (2017): [Winning team "AEGIS"](#). Presented a sustainable idea to empower refugees in Nepal using an e-commerce platform; presented at Hult Business School, Dubai.

Mentor: Global Venture Bootcamp, Asian Development Bank, Philippines — enhancing social ventures with emerging technologies.

Participant: IEEE AI & Big Data Workshop 2018; led multiple IEEE Student Chapter activities on campus.

Participant: UNICEF 1000 Golden Hours Code Camp (2017) — Best Prototype Award for mobile app idea.

References

Prof. Pål Halvorsen: Chief Research Scientist, SimulaMet; Professor, OsloMet; Adjunct Professor, UiO.
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